Note: Difference between e.m.f and potential difference:

As both e.m.f and potential difference are measured in volt, they may appear the same. But they are not. The e.m.f refers to the voltage developed across the terminals of an electrical source when it does not produce current in the circuit. Potential difference refers to the voltage developed between any two points (even across electrical devices) in an electric circuit when there is current in the circuit.

4.3 Electric circuit diagram

To represent an electrical wiring or solve problem involving electric circuits, the circuit diagrams are made.

The four main components of any circuits namely, (i) cell, (ii) connecting wire, (iii) switch and (iv) resistor or load are given above. In addition to the above many other electrical components are also used in an actual circuit. A uniform system of symbols has been evolved to describe them. It is like learning a sign language, but useful in understanding circuit diagrams. Some common symbols in the electrical circuit are shown in Table 4.1.

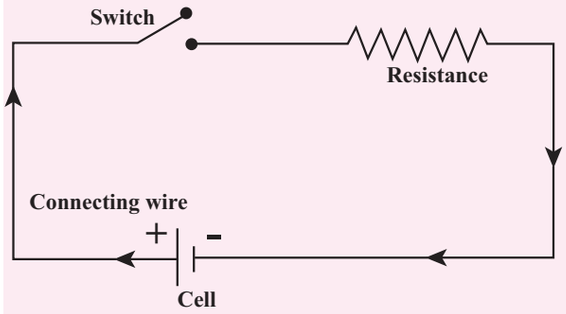


Figure 4.10 Typical electric circuit

Activity 1

Take a condemned electronic circuit board in a TV remote or old mobile phone. Look at the electrical symbols used in the circuit. Find out the meaning of the symbols known to you.

Table 4.1 Common symbols in electrical circuits

Symbol	Device	Symbol	Device	Symbol	Device
or	Switch		Wires joined	G or`	galvanometer
	Cell		Wires crossed	A	ammeter
	Battery		Fixed resistor	V	Voltmeter
	D .c . power` supply		variable resistor` (rheostat)		Two-way switch
	A.c. power supply		fuse		Earth connector
	Light bulb		Coil of wire		capacitor
	Potentiometer		transformer		thermistor
	light-depemdent resistor (LDR)		Semiconductor diode		bell

IX_SCI_EM_Unit-04_PHY.indd 44 12-02-2020 14:14:25